THE WRONSKIAN AND LINEAR INDEPENDENCE

Determining if a set of solutions to a differential equation is linearly independent

What is the Wronskian

Definition:

The Wronskian $W(\phi_1, \phi_2, \ldots, \phi_n)$ of n solutions $\phi_1, \ldots, \phi_n$ of a differential equation is the determinant of a matrix containing the functions and their successive derivatives:

$$W(\phi_1, \phi_2, \ldots, \phi_n) = \begin{vmatrix} \phi_1 & \phi_2 & \cdots & \phi_n \\ \phi_1' & \phi_2' & \cdots & \phi_n' \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1^{(n-1)} & \phi_2^{(n-1)} & \cdots & \phi_n^{(n-1)} \end{vmatrix}$$

Theorem:

If $\phi_1, \phi_2, \ldots, \phi_n$ are n solutions of $L(y)=0$ on an interval I, they are linearly independent there, if and only if,

$W(\phi_1, \phi_2, \ldots \phi_n)(x) \neq 0$ for all x in I.

If $W=0$, they linearly dependent.

Understanding the proof:-

We consider a linear combination of the solutions:

$$c_1\phi_1(x) + c_2\phi_2(x) + \ldots + c_n\phi_n(x) = 0$$

By taking derivatives, we get a system of n linear homogeneous equations:

$$c_1\phi_1'(x) + \ldots + c_n\phi_n'(x) = 0$$
$$\vdots$$
$$c_1\phi_1^{(n-1)}(x) + \ldots + c_n\phi_n^{(n-1)}(x) = 0$$

Since the determinant of this system is $W(\phi_1, \ldots, \phi_n)(x) \neq 0$, there is ONLY ONE trivial solution: $c_1 = c_2 = \ldots = c_n = 0$.
Thus, the functions are linearly independent.

Applications:-

1. Finding Fundamental Solution Sets:-
   Essential for finding the general solution of nth order linear homogeneous differential equations $L(y)=0$.
2. Checking Redundancy:-
   Determines if a set of functions contains linearly independent, non-redundant solutions.
3. Boundary Value Problems:-
   Crucial in verifying uniqueness, of solutions in physics & engineering (e.g. vibration, quantum mechanics.

How to Solve:-

1. Identify the functions:
   Gather the set of solutions $\phi_1, \ldots, \phi_n$.
2. Compute derivatives:
   Find successive derivatives up to order n-1.
3. Set up the Matrix:
   Place the functions in the first row, their 1st derivatives in the second row, up to $(n-1)^{th}$ derivatives in the last row.
4. Calculate the determinant:
   Evaluate the determinant of this $n \times n$ matrix.
5. Draw Conclusion:
   If $W(x) \neq 0$ on the interval, the functions are linearly independent. If $W(x) = 0$, they are linearly dependent.

Worked Example:-

Determine if $y_1 = e^{2x}$ and $y_2 = e^{3x}$ are linearly independent on any interval I.

1. Identify functions
   Let $\phi_1 = e^{2x}$ and $\phi_2 = e^{3x}$
2. Compute Derivatives
   $\phi_1' = 2e^{2x}$ and $\phi_2' = 3e^{2x}$
3. Set up the Wronskian
   $$W(e^{2x}, e^{3x}) = \begin{vmatrix} e^{2x} & e^{3x} \\ 2e^{2x} & 3e^{3x} \end{vmatrix}$$
4. Evaluate Determinant
   $$W = (e^{2x})(3e^{3x}) - (e^{3x})(2e^{2x})$$
   $$W = 3e^{5x} - 2e^{5x}$$
   $$W = e^{5x}$$
5. Analyse the Result:
   The value e^{5x} is never zero for any real $x \in \mathbb{R}$ ($e^{5x} \neq 0$).
6. Conclusion
   Since the Wronskian $W \neq 0$, the solutions e^{2x} and e^{3x} are linearly independent.

Name: Gawari Prashant Tanaji
Class: M.Sc. (Mathematics) Part I Sem I